VISHWAJIT MALL

Python Backend Developer | AI/LLM Engineer

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PROFESSIONAL SUMMARY

Python Backend Developer with **5+ years** of experience building production APIs, with hands-on AI/LLM engineering in production (SmartHR AI interview platform) and in 10+ open-source projects. Builds RAG pipelines (Qdrant, pgvector, hybrid search, citations), tool-calling and multi-agent systems with LangGraph and MCP, and LLM routing with fallback across OpenAI, Gemini, Groq, and local Ollama. Works daily with FastAPI, Pydantic, SQLAlchemy, Celery, Redis, PostgreSQL, Docker, and pytest. Published TraceNest on PyPI. Currently Backend Developer at SPNX.

TECHNICAL SKILLS

Languages: Python, SQL, TypeScript, JavaScript, PHP
Python Backend: FastAPI, async Python (asyncio), Pydantic v2, SQLAlchemy 2, Alembic, Celery, REST API design, GraphQL, WebSockets, OpenAPI/Swagger, microservices, OOP
AI / LLM Engineering: LLM integration (OpenAI, Gemini, Groq, Mistral, NVIDIA NIM, OpenRouter, Hugging Face, Ollama), RAG, LangChain, LangGraph, AI agents (ReAct, tool calling), multi-agent workflows, Model Context Protocol (MCP), LiteLLM, prompt engineering, embeddings, vector search (Qdrant, pgvector), LLM fallback routing
Databases & Vector Stores: PostgreSQL, MySQL, MongoDB, Redis, Qdrant, pgvector
Other Backend: Node.js, NestJS, Express.js, Prisma, Laravel
Security & Auth: JWT, OAuth 2.0 / OIDC (Keycloak, Microsoft Entra ID), RBAC, multi-tenant isolation, rate limiting, AES-256-GCM
Cloud & DevOps: Docker, Docker Compose, GitHub Actions, GitLab CI, AWS (S3, RDS, Amplify), Azure (Blob, Key Vault), Linux
Testing & Observability: pytest, pytest-asyncio, unit and integration testing, TraceNest, OpenTelemetry, Prometheus, Grafana
Tools & Practices: Git, GitHub, GitLab, JIRA, Azure DevOps, Agile/Scrum, system design, AI-assisted development (Claude Code)

PROFESSIONAL EXPERIENCE

Backend Developer | SPNX, India **Oct 2026 – Present**

FLO – VAT Registration Platform | *Node.js, NestJS 11, TypeScript, PostgreSQL 17, Prisma 7, Microsoft Entra ID, Azure, Jest*

- Sole backend developer (100 of 101 commits) of FLO, a staff workspace and client portal that runs EU VAT registrations for overseas businesses: **244 REST operations**, 34 NestJS modules, 63 Prisma models.
- Designed a data-driven workflow engine in pure TypeScript that runs each country's registration process as versioned data; the same guards enforce business rules and return each user's allowed actions.
- Implemented Microsoft Entra ID (OIDC) sign-in, Postgres-backed sessions with CSRF protection, and RBAC with 4 roles and 21 permissions; separate portal serializers keep internal data out of client responses.
- Cleared 3 security-test (VAPT) rounds; wrote 63 Jest spec files and a 24-suite API end-to-end harness; set up Claude Code project rules and skills for AI-assisted development.

Full Stack Developer | Bee Logical Software Solutions, Pune, India **Nov 2022 – Jun 2026**

- Core backend developer on 5 production platforms: designed APIs, data models, and auth, and worked with AI/ML engineers to ship LLM features. Received the **Outstanding Performance Award (Nov 2024)**.

SmartHR – AI Interview Platform | *Python, FastAPI, LangChain, LangGraph, RAG*

- Built an AI interview pipeline with LangChain for LLM orchestration and LangGraph for stateful, multi-turn interviews.
- Built RAG over job descriptions to generate role-specific questions, and agent-based scoring that writes candidate reports.
- Built the interview lifecycle APIs: session start, real-time state tracking, and result storage.

SetBot – Property Booking Platform | *Node.js, Express.js, TypeScript, MySQL, Angular*

- Built REST APIs for listings, bookings, and admin reports for studios, hostels, restaurants, and shops, with JWT auth and RBAC across 4 roles.

Bella Canvas – 3D Product Customization E-Commerce | *Laravel, Vue.js, MySQL (AWS RDS), AWS S3, Shopify API*

- Led backend design for a 5-stage product design lifecycle: REST APIs for designs and assets, Shopify order sync, and an S3 pipeline serving textures for real-time 3D rendering.

MahaEseva.com – Online Rent Agreements | *Laravel, MySQL*

- Built admin and customer portals to create and e-sign government-compliant rent agreements across India, with online payments, biometric verification, RBAC, and lead management.

Havu – Events-Based Dating Platform (Admin Portal) | *React, GraphQL, AWS Amplify*

- Built the admin portal for events, users, cities, and venues, with RBAC and multi-city scheduling.

Software Developer | WebReinvent, New Delhi, India **Jul 2022 – Oct 2022**

- Built features and fixed bugs in VaahCMS, an open-source Laravel and Vue.js CMS (MySQL, REST APIs).

Full Stack Developer | **BluBrandz Technologies Pvt Ltd, New Delhi, India** **Aug 2021 – Jun 2022**

- Built Angle Fertility (hospital management: patient records, appointments, clinical workflows), Aragma (event management), and Unicorn Equipments (equipment tracking dashboard) with Laravel and MySQL.

PROJECTS AND OPEN SOURCE

DevFlow-AI – AI Code Review and CI/CD Platform | github.com/vishwajitvm/DevFlow-AI

Tech: *Python 3.13, FastAPI, SQLAlchemy 2, Celery, PostgreSQL, Redis, MCP, Ollama, Next.js 16, Docker, OpenTelemetry*

- Built a self-hosted AI code-review, auto-fix, and CI/CD platform for GitHub and GitLab: **132 API endpoints**, 31 tables, 30 Celery tasks, 17 Docker services.
- Wrote a framework-free tool-calling agent loop for 9 roles (analyst, fixer, code and security reviewers) with JSON-schema-validated output, token budgets, and AST caller-compatibility checks.
- Built an LLM router for 90 free models (10 providers) with Retry-After-based 429 cool-downs and local Ollama fallback.
- Routed every GitHub/GitLab write through one **MCP gateway** (57 allow-listed tools) with one-time keys and a deny-by-default go / ask / stop rules engine.
- Ran reviews in no-network, read-only Docker sandboxes; **392 unit, 67 API end-to-end, and 43 Playwright tests** run in GitHub Actions and GitLab CI.

OmniShell – LLM-Planned Shell Automation with Human Approval | github.com/vishwajitvm/OmniShell

Tech: *Python, FastAPI, LiteLLM, Pydantic, PostgreSQL, Redis, NestJS, psutil, Docker*

- Built a FastAPI service that turns natural-language requests into OS workflows (Linux, macOS, Windows) across 19 capability types, using regex intent routing and one LiteLLM planning call with Pydantic-validated JSON.
- Implemented LLM fallback across 6 providers (Mistral, OpenRouter, Groq, Gemini, NVIDIA NIM, Hugging Face) with error classification and per-model cool-downs.
- Designed a PostgreSQL scheduler with FOR UPDATE SKIP LOCKED claims, heartbeat leases, and SHA-256-hashed expiring approval tokens for human sign-off.

RepoMind – AI Codebase Q&A with RAG and LangGraph | github.com/vishwajitvm/RepoMind

- Built a LangGraph and FastAPI assistant that answers questions about GitHub repositories with file and line citations, using AST-based code chunking, Qdrant vector search, live GitHub API lookups, and 9-provider LLM fallback.

TenantMind-AI – Multi-Tenant AI Property Management | github.com/vishwajitvm/TenantMind-AI

- Built a multi-tenant FastAPI backend (database per tenant in MongoDB, Keycloak JWT), per-tenant Qdrant collections for document search, Celery workers, and an LLM gateway with fallback (Gemini, Groq, OpenRouter, Ollama) that logs latency and tokens per call.

AgentHive – Multi-Agent Orchestration Platform | github.com/vishwajitvm/AgentHive

- Built a ReAct agent platform (FastAPI, PostgreSQL, Celery, Next.js) with a database-configured LLM router (Gemini, OpenAI, Groq, Hugging Face, Ollama) with retry and fallback, 40 tools with per-agent allow-lists, and pgvector memory.

PolicyBot – RAG Q&A for Company Policies | github.com/vishwajitvm/PolicyBot

- Built a hybrid RAG pipeline (FastAPI, Qdrant, MongoDB, React) with vector and keyword retrieval, freshness-aware ranking, citations, and confidence scores, a LangGraph ingestion workflow, and per-stage traces for evaluation.

TraceNest – Python Logging SDK on PyPI | pypi.org/project/tracenest

- Published a zero-dependency, thread-safe logging SDK (18 releases) with daily rotation, recursive secret masking, FastAPI middleware, a built-in log viewer, and 42 tests; used in DevFlow-AI and OmniShell.

Tallyko – Multi-Tenant Restaurant and Retail POS | github.com/vishwajitvm/Tallyko

- Built a multi-tenant POS (FastAPI, PostgreSQL row-level security, 38 tables, Redis) with an offline-first React Native (Expo) app that syncs queued bills without duplicates, local OCR menu import, and 400+ automated tests.

Hopline – Encrypted Peer-to-Peer File Transfer | github.com/vishwajitvm/Hopline

Tech: *Node.js, WebRTC, WebSockets, React 19, Vite, Electron, AES-256-GCM, Docker, GitHub Actions*

- Built a peer-to-peer file transfer app for browser, desktop (Electron: Windows, macOS, Linux), and CLI with a Node.js WebSocket signaling server; live at hopline.onrender.com.
- Built encrypted TCP transfer for CLI and desktop (CPace PAKE, HKDF, AES-256-GCM): **10 GiB in 56.8 s**, SHA-256 verified.

More AI Projects | github.com/vishwajitvm

- **LokTathya**: civic data platform for India (FastAPI, Next.js, PostgreSQL/PostGIS, 50 tables) with a GitHub Actions pipeline running migrations and 76 tests.
- **NL-Automation-Platform**: plain-English automations on 6 FastAPI microservices, with a LangGraph clarification agent and a fail-closed safety classifier.
- **FitTrack-AI**: meal nutrition from a photo via vision LLMs, grounded in USDA and Open Food Facts data (Next.js, FastAPI).

EDUCATION

Bachelor of Science (B.Sc.) | Panjab University, Chandigarh **2015 – 2018**